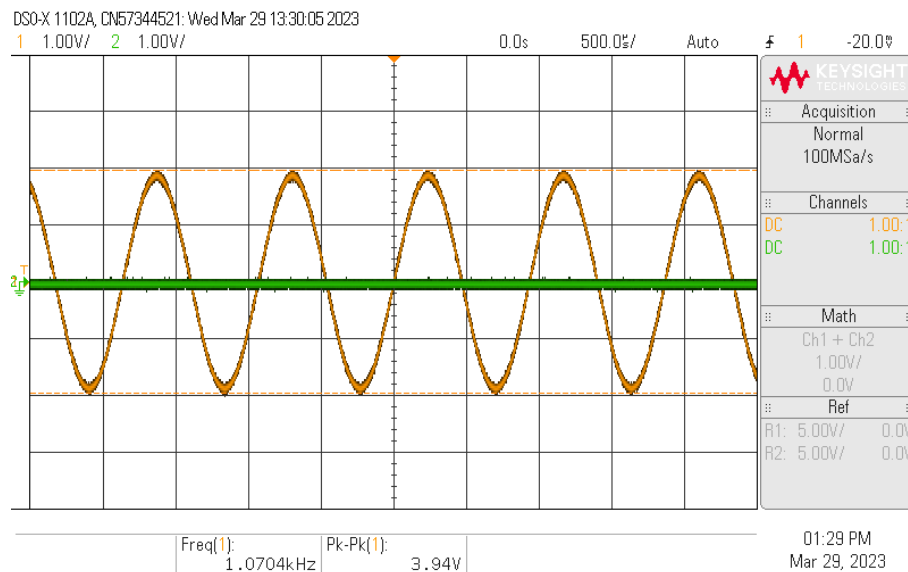


Q1.



Q2.

$$V_3 = -2.0356V$$

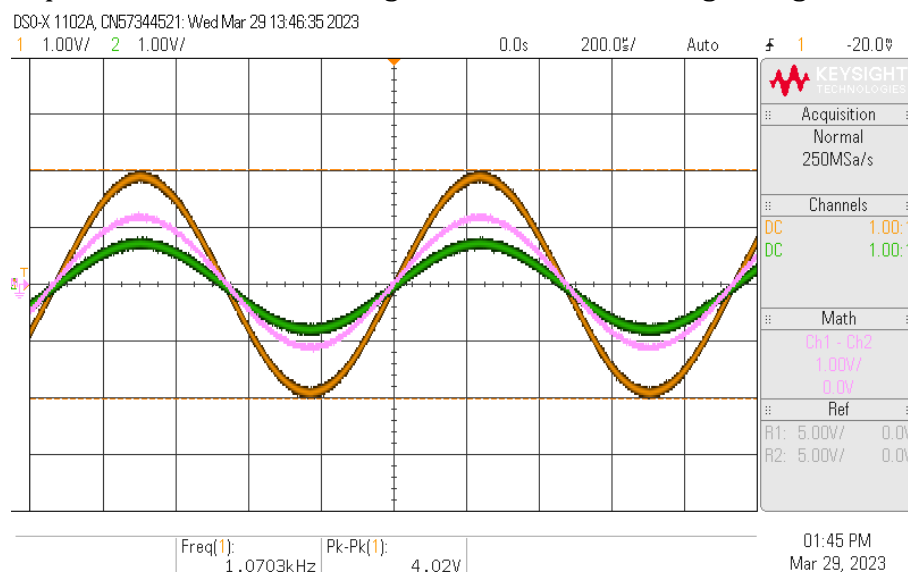
It is negative because the polarity of the voltage is flipped around. The positive sign is pointing at ground.

Q3.

$$I_2 = 12.948mA$$

Q4.

We cannot directly measure the voltage across R_1 because it is not directly connected to ground. Therefore, CH1 is used to measure the signal generator waveform and CH2 is used to measure the voltage across R_2 and R_3 . The math function (purple curve) is utilised to display CH1 – CH2 on the oscilloscope which represent the voltage across R_1 . The negative terminal (black clip) of the oscilloscope must be connected to ground when measuring voltages.



Q5.

We use 'amplitude' to measure the peak-to-peak voltage of the square wave here because 'peak-to-peak' measurement on the oscilloscope also takes into account the overshoot and spikes on the square wave (if you observe carefully on the edges of the square wave, there are small spikes). However, **this is not required knowledge for this course**. You just need to know how to measure and display the capacitor (and inductor) voltage at the correct edge and how to set the trigger level and the signal generator at the correct amplitude and frequency.

